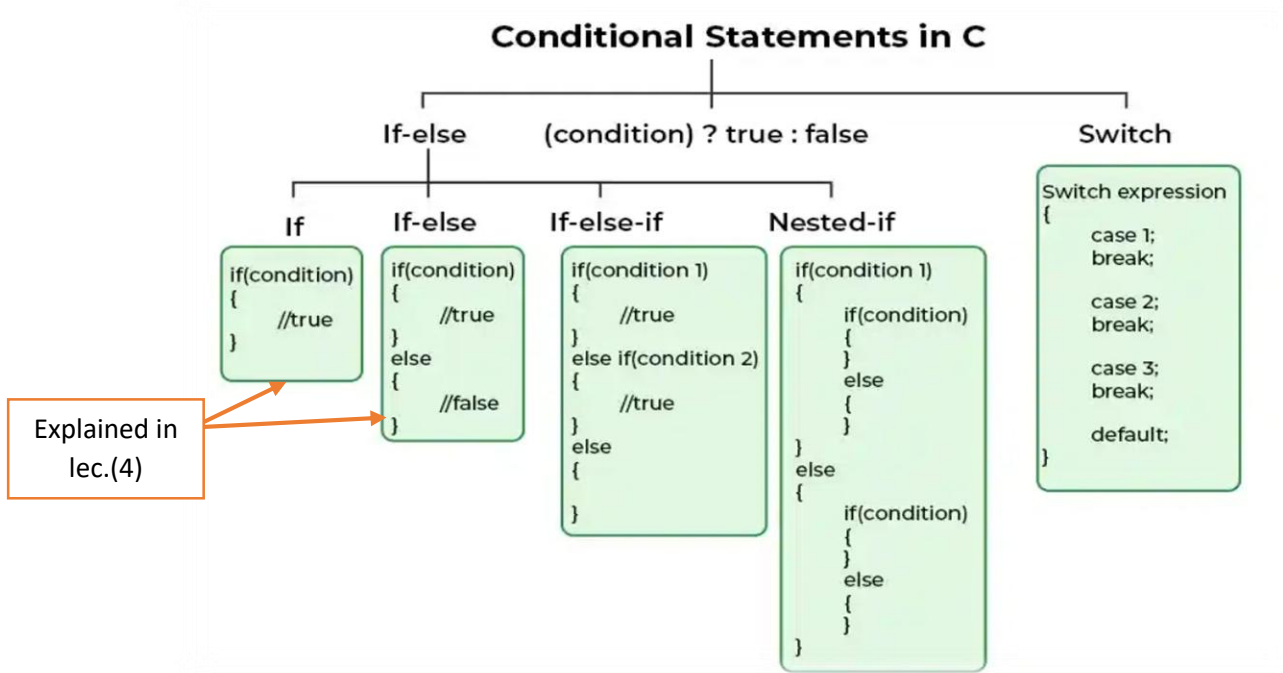


Types of Conditional Statements in C++



5-1 The if else if Statement

The [if else if statements](#) are used when the user has to decide among multiple options. The C++ if statements are executed from the top down. As soon as one of the conditions controlling the if is true, the statement associated with that if is executed, and the rest of the C else-if ladder is bypassed. If none of the conditions is true, then the final else statement will be executed.

Example (1): Write a program that allows those aged 20 to vote; if they are less than or equal to 10 years, they are prevented from voting; if they are a teen, display an explanatory message about the waiting years; if they are older, prevent them from voting.

Sol.

```

#include<iostream>
using namespace std;
int main()
{
    int i ;
    cout<< "please input your age as i = "<<endl;
    cin>>i;
    // If else ladder with three conditions
    if (i <= 10)
        cout<<"Not Eligible";
    else if (i == 15)

```

Output:

please input your age as i = 20

You can vote

```

    cout<<"wait for five years";
else if (i == 20)
    cout<<"You can vote";
else
    cout<<"Not a valid age";

    return 0;
}

```

Example (2): Write a C++ program to read a number and print the day of the week:

Sol.

```

#include<iostream>
using namespace std;
int main()
{
    int day;
    cout<< "please input the number of day ::";
    cin >> day;
    if ( day == 1 ) cout << "Sunday";
    else if (day == 2 ) cout << "Monday";
    else if (day == 3 ) cout << "Tuesday";
    else if (day== 4) cout << "Wednesday";
    else if (day== 5) cout << "Thursday";
    else if (day == 6 ) cout << "Friday";
    else if (day == 7 ) cout << "Saturday";
    else cout << "Invalid day number";
}

```

The output:
please input the number of day :: 4
Wednesday

Example (3): Write a C++ program to compute the value of Z according to the following equations:

$$Z = \begin{cases} x = x + 5 & : x < 0 \\ \cos(x) + 4 & : x = 0 \\ \sqrt{x} & : x > 0 \end{cases}$$

Sol.

```

#include<iostream>
#include <cmath>
using namespace std;
int main()
{
    int Z, x;
    cout<<"Enter X value";
}

```

cmath: library has many functions that allow you to perform mathematical tasks on numbers. (sin, cos ,abs, log ,exp, sqrt,.....)

```
cin>> x;
if ( x < 0 ) {Z= x + 5;}
else if ( x == 0 ) {Z= cos(x) + 4;}
else {Z= sqrt(x);}
cout << "Z is " << Z;
}
```

The output:
Enter X value 25
Z is 5

5-2 The nested if Statement

- ✚ A **nested if statement** in C++ refers to an `if` statement placed inside another `if`, `else if`, or `else` block. This allows you to check for a second, inner condition only if the outer condition has already been met.
- ✚ We use nested if-else statements when we want to implement multilayer conditions (condition inside the condition inside the condition and so on).

Example (1): Write a C++ program to read a number and check it if the remainder is zero when we divide the number by 2 and 3:

Sol.

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cout<<"please, input the number n: ";
    cin >> n;

    // Outer if statement
    if (n % 2 == 0) {

        // Inner if statement
        if (n % 3 == 0) {
            cout << "Divisible by 2 and 3";
        }
        else {
            cout << "Divisible by 2 but not 3";
        }
    }
    else {
        cout << "Not Divisible by 2";
    }
}
return 0;
}
```

The output:
please, input the number n: 6
Divisible by 2 and 3

5-3 Switch selection statement

The `switch` statement in C++ is a **multi-way selection mechanism** that provides an alternative to a long chain of `if-else-if` statements. It allows a program to execute different blocks of code based on the value of a single expression.

Syntax:

The basic syntax of a switch statement is as follows:

```
switch (expression) {  
    case value1:  
        // code to execute if expression equals value1  
        break;  
    case value2:  
        // code to execute if expression equals value2  
        break;  
    ...  
    default:  
        // code to execute if expression does not equal any case  
}
```

Example (1): Write program to print name of weekday match to its number.

Sol.

```
#include <iostream>  
using namespace std;  
int main()  
{  
    int day = 4;  
  
    // Determine the day name using switch  
    switch (day)  
    {  
        case 1:  
            cout << " Sunday";  
            break;  
        case 2:  
            cout << " Monday";  
            break;  
        case 3:  
            cout << "Tuesday";  
            break;  
        case 4:  
            cout << "Wednesday";  
            break;  
        case 5:  
            cout << "Thursday";  
            break;  
        case 6:
```

```
    cout << "Friday";  
    break;  
case 7:  
    cout << "Saturday";  
    break;  
default:  
    cout << "Invalid day number";  
    break;  
}  
return 0;  
}
```

Output:

Wednesday

Homework:

- 1) Develop the program above (**with switch statement**) to display the value of degrees for one student based on the range of degree in one subject. By benefited from the following table.

The degree	The value
≥ 90	Excellent
$\geq 80 \ \& \ < 90$	Very good
$\geq 70 \ \& \ < 80$	good
$\geq 60 \ \& \ < 70$	accepted
< 60	Fail

- 2) Write a program to read X and print Sin X if $X > 0$, square root X if $X < 0$ and absolute X if $X/2$ is integer.
- 3) Write C++ program to read 5 numbers and determine if the numbers sorted ascending or not.

5-4 loop statements

- ✚ C++ provides several types of loop statements to repeatedly execute a block of code until a specific condition is met.

5-4-1 for Loop

The **for loop** is an **entry-controlled loop**, which means that it checks whether the test condition is true before executing the statements inside it.

Syntax

```
for (initialization; condition; updating) {  
    // body of for loop  
}
```

The various **parts of the for loop** are:

- **Initialization:** Initialize the variable to some initial value.
- **Test Condition:** This specifies the test condition. If the condition evaluates to true, then body of the loop is executed. If evaluated false, loop is terminated.
- **Update Expression:** After the execution loop's body, this expression increments/decrements the loop variable by some value.

All these together is used to create a logic and flow of the loop.

Example (1): using for loop, write program to print the number from 1 to 5.

Sol

```
#include <iostream>  
using namespace std;  
int main() {  
    // For loop that starts with i = 1 and ends when i is greater than 5.  
    for (int i = 1; i <= 5; i++)  
    {  
        cout << i << " ";  
    }  
    return 0;  
}
```

Output:

1 2 3 4 5

Develop this example to print a
countdown from 5 to 1:

Example (2): using *for loop*, write program to print the even number between 0 to 10.

Sol.

```
#include <iostream>
using namespace std;
int main() {

    for (int i = 0; i <= 10; i = i + 2)
    {
        cout << i << "\n";
    }
    return 0;
}
```

Output:

0
2
4
6
8
10

5-4-2 while Loop

- ✚ The **while loop** is also an **entry-controlled loop** which is used in situations where we do not know the exact number of iterations of the loop beforehand.
- ✚ In for loop, we have seen that the number of iterations is **known beforehand**, i.e. the number of times the loop body is needed to be executed is known to us and we create the condition on the basis of it. But while loops execution is solely based on the condition.

Syntax

```
while (condition) {
    // Body of the loop
    // update expression
}
```

Only the **condition** is the part of while loop syntax, we have to **initialize** and **update loop** variable manually.

Example (1) : using while loop, write program to print the number from 1 to 5.

```
#include <iostream>
using namespace std;
int main() {
    // Initialization
    int i = 1;
    // while loop that starts with i = 1 and ends when i is greater than 5.
    while (i <= 5)
    {
        cout << i << " ";
        // Updating
        i++;
    }
    return 0;
}
```

Output:

1 2 3 4 5

Example (2) : Write C++ program to find the summation of the following series using while loop:

$$\sum_{i=1}^n i^2 = 1^2 + 2^2 + 3^2 + \dots n^2$$

Sol.

```
#include <iostream>
using namespace std;

int main() {

    int i = 1, n ,sum = 0;
    cout << "enter positive number";
    cin >> n;
    while ( i <= n )
        {
            sum+= i * i;
            i++;
        }
    cout << "sum is:"<< sum<< endl;

    return 0;
}
```

Output:

enter positive number 4

sum is: 30

Homework:

- 1) **Using while loop**, write a C++ program to find the summation of student's marks, and its average, assume the student have 8 marks.
- 2) **Using while loop**, write a C++ program to find the summation of the odd numbers, between 0 and 100)